Series

Section 9.2-9.6

2-18-26

A) Does $\sum a_n$ Converge?

$\sum a_n$

$\left\{\frac{1}{n}\right\} = \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\right\}$

$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$

PARTIAL SUMS

$S_1 = 1$

$S_2 = 1 + \frac{1}{2} = \frac{3}{2}$

$S_3 = 1 + \frac{1}{2} + \frac{1}{3} = \frac{11}{6}$

$S_4 = \frac{11}{6} + \frac{1}{4} = \frac{25}{12}$

$\left\{1, \frac{3}{2}, \frac{11}{6}, \frac{25}{16}, \dots\right\}$

Diverges

B) The Tests

1) Geometric Series

$|r| \geq 1$ divergent

$|r| \leq 1$ convergent

Finite Sum	Infinite Sum
$S_n = \frac{a_1(1-r^n)}{1-r}$	$S_\infty = \frac{a_1}{1-r}$

Ex: $\sum_{1}^{\infty} 3\left(\frac{1}{4}\right)^n$

$\left|\frac{1}{4}\right| \leq 1$ converges by geometric series test

$S_\infty = \frac{\frac{3}{4}}{1-r}$

$\sum_{0}^{\infty} \frac{3}{10}\left(\frac{1}{10}\right)^n$ $\left|\frac{1}{10}\right| \leq 1$

c by geo test

$S_\infty = \frac{3}{9}$

2) Telescoping (partial fractions)

When the sequence collapses in on itself

$\sum_{1}^{\infty} \frac{1}{n(n+2)}$ $\frac{A}{n} + \frac{B}{n+2} = \frac{1}{n(n+2)}$

$A(n+2) + Bn = 1$

$A = \frac{1}{2}$ $B = -\frac{1}{2}$

$\sum_{1}^{\infty} \frac{1}{2}\left(\frac{1}{n} - \frac{1}{n+2}\right)$

$\frac{1}{2}\left(\left(1 - \cancel{\frac{1}{3}}\right) + \left(\frac{1}{2} - \cancel{\frac{1}{4}}\right) + \left(\cancel{\frac{1}{3}} - \frac{1}{5}\right) + \left(\cancel{\frac{1}{4}} - \frac{1}{6}\right)\right)$

$\frac{1}{2}\left(1 + \frac{1}{2}\right) - \lim_{n\to\infty}\left(\frac{1}{n+2}\right) = \frac{1}{2}\left(\frac{3}{2}\right)$

Converging by tele

Ex: $\sum_{1}^{\infty}\left(\frac{1}{2n+1} - \frac{1}{2n+3}\right)$

$\left(\left(\frac{1}{3} - \cancel{\frac{1}{5}}\right) + \left(\cancel{\frac{1}{5}} - \frac{1}{7}\right) \dots\right)$

$\left(\frac{1}{3} - \lim_{n\to\infty}\left(\frac{1}{2n+3}\right)\right) = \frac{1}{3}$

converging by telescope

3) nth term test

$\lim_{n\to\infty} a_n \neq 0 \to$ Diverges

$\lim_{n\to\infty} a_n = 0 \longrightarrow$ Fail

Ex: $\sum_{1}^{\infty} \frac{n^2}{2n^2+1}$

$\lim_{n\to\infty} \frac{n^2}{2n^2+1} = \frac{1}{2} \neq 0$

Diverges by nth term

Ex: $\sum_{1}^{\infty} \frac{2^n+1}{2^{n+1}} = \sum_{1}^{\infty} \frac{2^n+1}{2\cdot 2^n}$

$\sum_{1}^{\infty} \frac{2^n}{2\cdot 2^n} + \sum_{1}^{\infty} \frac{1}{2\cdot 2^n} = \sum_{1}^{\infty} \frac{1}{2}\left(\frac{1}{2}\right)^n$

$\lim_{n\to\infty} \frac{1}{2} = \frac{1}{2} \neq 0$ + $\frac{1}{2} \leq 1$

Diverge convergo

Diverges by nth term

4) Integral Test

$f(x)$ = general term

Conditions: $f(x)$ is cont., positive, decreasing

if $\int_a^\infty f(x)\,dx$ converges

then $\sum a_n$ Converges

if $\int_a^\infty f(x)\,dx$ diverges

then $\sum a_n$ diverges

Ex: $\sum_{2}^{\infty} \frac{\ln(n)}{n}$ $f(x) = \frac{\ln x}{x}$

$f(x)$: cont. ✓ positive ✓ decreasing ✓

$\int_2^\infty \frac{\ln x}{x}\,dx = \lim_{a\to\infty} \left.\frac{(\ln x)^2}{2}\right|_2^a$

$\lim_{a\to\infty}\left(\frac{1}{2}(\ln a)^2 - \frac{1}{2}(\ln 2)^2\right)$ ↳ diverges

$\boxed{\sum a_n \text{ diverges by integral test}}$

Ex: $\sum_{1}^{\infty} \frac{1}{n}$ $f(x) = \frac{1}{x}$

$f(x)$ is: cont. ✓ positive ✓ decreasing ✓

$\int_1^\infty \frac{1}{x}\,dx = \lim_{a\to\infty} \ln|x|\Big|_1^a$

$= \lim_{a\to\infty}(\ln a - \ln 1) = \infty$

$\boxed{\sum a_n \text{ diverges by integral test}}$

Ex: $\sum_{2}^{\infty} \frac{1}{n^2}$

$\lim_{a\to\infty} -\frac{1}{x}\Big|_2^a$

$\boxed{\text{converges to } \frac{1}{2}}$

5) P-series

$\sum_{1}\left(\frac{1}{n}\right)^p$ $p > 1$ $\sum a_n$ converges

$p \leq 1$ $\sum a_n$ diverges

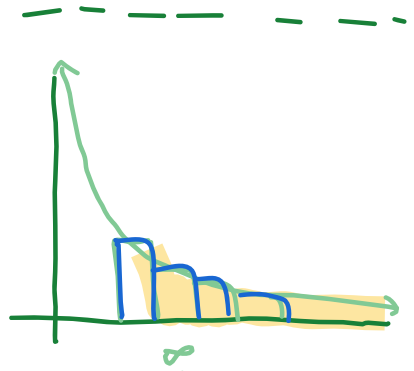
Ex: $\sum\left(\frac{1}{n}\right)^2$

$\sum a_n$ converges

Ex: $\sum\left(\frac{1}{\sqrt{n}}\right)$

$\sum a_n$ diverges

Ex: $\sum_{2}^{\infty} \frac{1}{n^2}$



$\int_2^\infty \frac{1}{x^2}\,dx < \sum a_n < \int_2^\infty \frac{1}{x^2}\,dx$ + 1st term

$\frac{1}{2} < \sum a_n < \frac{1}{2} + \frac{1}{4}$

$\frac{1}{2} < \sum a_n < \frac{3}{4}$

6) Direct Comparison

$\sum_{1} b_n \leftarrow$ sum that you know

$\sum_{1} b_n$ to conv

$\sum_{1} a_n \leq \sum_{1} b_n$, $\sum_{1} a_n$ converges

$\sum_{1} b_n$ to diverge

$\sum_{1} b_n \leq \sum a_n$ diverges

Ex: $\sum_{1}^{\infty} \frac{3n}{n^3+4n}$ $\sum_{1}^{\infty} \frac{3n}{n^3} = \sum_{1}^{\infty} \frac{3}{n^2}$

$P = 2 > 1$

$\sum b_n$ converges by p-series

$\frac{3}{n^2+4n} \leq \frac{3}{n^2}$

$3n^3 \leq 3n^3 + 12n$

$\sum a_n$ converges by direct comparison

Ex: $\sum_{1}^{\infty} \frac{1+n\ln n}{n^2+5}$

$\sum_{1}^{\infty} \frac{n\ln(n)}{n^2} = \sum_{1}^{\infty} \frac{\ln(n)}{n}$

$f(x) = \frac{\ln x}{x}$

cont, positive, decreasing

$\int_1^\infty \frac{\ln x}{x}\,dx = \lim_{a\to\infty} \frac{1}{2}(\ln x)^2\Big|_1^a$

$\lim_{a\to\infty} \frac{1}{2}(\ln a)^2 - 0 = \infty$

$\sum b_n$ diverges by integral test

$\frac{\ln(n)}{n} \leq \frac{1+n\ln(n)}{n^2+5}$

$\cancel{n^2\ln(n)} + 5\ln(n) \leq n + \cancel{n^2\ln(n)}$

$5\ln(n) \leq n$

$\boxed{\sum a_n \text{ diverges by direct comparison}}$

Ex: $\sum_{1}^{\infty} \frac{2n+1}{n^2+2n+1}$

$\sum_{1}^{\infty} \frac{2}{n}$

$P = 1 = 1$ $\sum b_n$ diverges by P-series

$\frac{2}{n} \leq \frac{2n+1}{n^2+2n+1}$

$2n^2 + 4n + 2 \leq 2n^2 + n$

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Direct Comparison doesn't work

Using Limit Comparison

$\lim_{n\to\infty} \frac{\frac{2n+1}{n^2+2n+1}}{2/n} = 1$

1 is finite, non-zero

$\sum a_n$ diverges by limit comparison

7) Limit Comparison

$\lim_{n\to\infty} \frac{a_n}{b_n}$ = finite, non-zero #

↳ $\sum a_n$ and $\sum b_n$ do the same.

$\lim_{n\to\infty} \frac{a_n}{b_n} = 0$

↳ $\sum b_n$ conv.	$\sum b_n$ div.
$\sum a_n$ conv.	test fails

Ex: $\sum_{1}^{\infty} \frac{1}{2^n+1}$

$\sum b_n = \sum_{1}^{\infty}\left(\frac{1}{2}\right)^n$

$\frac{1}{2} \leq 1$ converges by geometric

$\frac{1}{2^n+1} \leq \left(\frac{1}{2}\right)^n$

$2^n \leq 2^n + 1$ ✓

$\boxed{\sum a_n \text{ conv. by direct comp.}}$

8) Ratio Test

$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| < 1$

converges

$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| > 1$

Diverges

$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = 1$

Test fails

Ex: $\sum_{1}^{\infty} \frac{1}{n!}$

$\lim_{n\to\infty}\left|\frac{\frac{1}{(n+1)!}}{\frac{1}{n!}}\right| = \lim_{n\to\infty}\left|\frac{n!}{(n+1)!}\right|$

$= \lim_{n\to\infty}\left|\frac{1}{n+1}\right| = 0 < 1$

$\sum a_n$ conv by ratio test

Ex: $\sum_{1}^{\infty} \frac{n^n}{n!}$

I believe this diverges

Ex: $\sum_{1}^{\infty} \frac{n^2(-1)^n}{3^n}$

$\lim_{n\to\infty}\left|\frac{(n+1)^2\cancel{(-1)^{n+1}}}{3^{n+1}} \cdot \frac{\cancel{3^n}}{n^2\cancel{(-1)^n}}\right|$

$\lim_{n\to\infty}\left|\frac{(n+1)^2}{3n^2}\right| = \frac{1}{3} < 1$

$\boxed{\sum a_n \text{ conv. by ratio test}}$

Ex: $\sum_{1}^{\infty} \frac{n^5}{(2n)!}$

$\lim_{n\to\infty}\left|\frac{(n+1)^5}{(2n+2)!} \cdot \frac{(2n)!}{n^5}\right|$

$\lim_{n\to\infty}\left|\frac{1}{(2n+2)(2n+1)}\right|$

9) Root Test

$\lim_{n\to\infty} \sqrt[n]{|a_n|}$

< 1 converges

> 1 diverges

$= 1$ Fails

10) Alternating Series Test

$(-1)^n$ $(-1)^{n+1}$

① $\sum a_n$ is alternating

② $\lim_{n\to\infty} a_n = 0$

③ $a_{n+1} < a_n$

$|\text{error}| < |\text{next term}|$

Ex: $\sum_{1}^{\infty} \frac{(-1)^n}{n}$ ① $\sum a_n$ is alternating

② $\lim_{n\to\infty} \frac{1}{n} = 0$

③ $\frac{1}{n+1} < \frac{1}{n}$

$n < n+1$

$\boxed{\sum a_n \text{ is converging by alternating series}}$

Ex: $\sum_{1}^{\infty} \frac{(-1)^{n+1}(n+2)}{n(n+1)}$

① $\sum a_n$ is alternating

② $\lim_{n\to\infty} \frac{n+2}{n^2+n} = 0$

③ $\frac{n+3}{(n+1)(n+2)} < \frac{n+2}{n(n+1)}$

$n\cancel{(n+1)}(n+3) < (n+2)^2\cancel{(n+1)}$

$\cancel{n^2} + 3n < \cancel{n^2} + 4n + 4$

$\sum a_n$ is converging by alternating series

Partial Sum w/ 5 terms

$\sum_{1}^{\infty} \frac{(-1)^n}{n}$

$S_5 = -1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5}$

$|\text{error}| < |\text{next term}|$

$|\text{error}| < \frac{1}{6}$

$S_5 - \frac{1}{6} < S_\infty < S_5 + \frac{1}{6}$

Ex: How many terms needed to approx

$\sum_{0}^{\infty} \frac{(-1)^n}{n!}$ so error $< \frac{1}{10000}$

$\sum a_n$ is alternating

$\lim_{n\to\infty} \frac{1}{n!} = 0$ $\frac{1}{(n+1)!} < \frac{1}{n!}$

$n! < (n+1)!$

$\sum a_n$ conv by alternating series test

$|\text{error}| < |\text{next term}|$

$1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120} + \frac{1}{720} - \frac{1}{5040} + \frac{1}{40320}$

$\boxed{8 \text{ terms}}$

Cool Vocab

(alternating series)

Absolutely Convergent	Conditionally Convergent
$\sum_{1}^{\infty} \frac{(-1)^n}{n^2}$	$\sum_{1}^{\infty} \frac{(-1)^n}{n}$
$\lim_{n\to\infty} \frac{1}{n^2} = 0$	$\sum a_n$ is alt
$\frac{1}{n+1} < \frac{1}{n}$	$\sum a_n$ conv. alternating series
$n < n+1$ ✓	$\lim_{n\to\infty} \frac{1}{n} = 0$
$\sum a_n$ converging by alternating series test	$\sum a_n$ conv. by alternating series
$\sum_{1} \lvert a_n \rvert$	$\sum_{1}^{\infty} \frac{1}{n}$
$\sum_{1}^{\infty} \frac{1}{n^2}$	$P = 1 = 1$
$P = 2 > 1$	Divergent by p-series
$\sum \lvert a_n \rvert$ converges	$\boxed{\text{conditionally convergent}}$
$\boxed{\text{absolutely convergent}}$	